

Variational Quantum Eigensolver Simulations for Multi-Qubit Ising Models

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Abstract

Variational quantum eigensolvers (VQEs) are hybrid quantum-classical algorithms utilizing parameterized quantum circuits and classical optimization cost functions to determine the ground state of a given physical system. In this project, variational quantum eigensolvers were utilized to determine the ground state energy and magnetization of transverse field Ising models. We began by determining the most efficient ansatz for the variational quantum eigensolver. Efficiency was measured through comparison of the average number of iterations before convergence and average percent error per ansatz. Then, the variational quantum eigensolver was applied to a two-qubit and four-qubit Ising model. The accuracy of the variational quantum eigensolver solution was compared to a classically-determined solution (through linear algebra methods). Further investigations seek to study the use of variational quantum eigensolvers for more complex spin models with different particle interactions.

Index Terms: variational quantum eigensolver, Ising model, quantum computing.

1 Introduction

This project was completed through the Next Generation Pathways to Computing Internship at Oak Ridge National Laboratory. This program seeks to introduce high school students to professional research at a national laboratory.

This paper requires a background in basic quantum theory and students are recommended to acquire the given background before continuing. A recommended resource for acquiring the necessary background is the free IBM Quantum Learning course on an introduction to quantum information (Watrous, 2025a). The majority of this paper is directed at those with limited knowledge in quantum computation.

1.1 Paper Organisation

We begin with a discussion of the necessary background knowledge for the project. A brief introduction to variational quantum eigensolvers and transverse field Ising models is given. Then, a description of the methods follows. Within this description, the relevant findings are included. Finally, an analysis of the results is provided along with suggestions for future research.

2 Background

The following background is directed at an undergraduate audience with very elementary understanding of quantum theory.

2.1 Transverse Field Ising Model

The Ising Model is a mathematical model that describes ferromagnetism in chemistry. It traditionally assumes that there is a lattice of particles, each with an up or down spin. The spins interact and the system tends to a ground state with the lowest energy possible. This model has already been solved in numerous configurations. However, this traditional model is a completely classical system. By applying a magnetic field to the model, it introduces quantum properties. Below is a description of the particular model at the center of this project.

Consider a transverse field one-dimensional Ising model. In this model, the particles interact with one another isotropically (no

one direction is favored over another) and only nearest-neighbor interactions are considered. We derive the relevant Hamiltonian equation from this model (A Hamiltonian is an equation that describes the total energy of a system). The general form for a one-dimensional Ising model Hamiltonian is:

$$\hat{H} = -J \sum_{\langle i \rangle} Z_i \otimes Z_{i+1} + h \sum_i X_i \quad (1)$$

where Z and X represent the Z - and X - Pauli operations, respectively. Within the first sum, tensor products are utilized. The coefficient J represents the strength of the interaction between particles and the coefficient h represents the field strength. As the equation describes, the interactions between particles occur in the Z -direction and the external field is exerted in the X -direction. In all systems considered, J is arbitrarily set to 1.

2.1.1 Two-Qubit Model Consider a two-qubit Ising model (See Figure 1). The particles are three dimensional, so the actual spins are a superposition of up and down spins. The Hamiltonian for this system is:

$$\hat{H} = -JZ_1 \otimes Z_2 + h(X_1 + X_2) \quad (2)$$

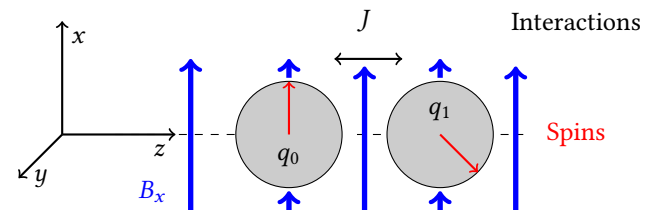


Figure 1. Two Qubit Ising Model

There are a few notable states of this system as we vary the field imposed. If no field is imposed, the system tends to a ferromagnetic state, where both spins are upward (See Figure 2). In this case, $h = 0$ and the Hamiltonian can be written $\hat{H} \approx -Z_1 Z_2$ (since $J = 1$).

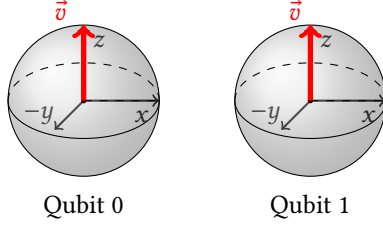


Figure 2. Two-Qubit Ising Model: Ferromagnetic State

If a very strong field is imposed (such that $h \gg J$), the system tends to a paramagnetic state, where both spins are in the positive X-direction (See Figure 3). Since $h \gg J$, the Hamiltonian can be written $\hat{H} \approx h(X_1 + X_2)$. A symmetric result is found for $h \ll -J$ where the spins point in the negative X-direction.

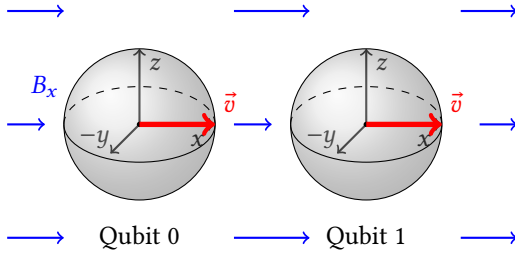


Figure 3. Two-Qubit Ising Model: Paramagnetic State

Between these states, when $H \approx J$, a phase transition occurs. In this case, the spins are in random positions in each particle as the interactions and external field fight for dominance in the system (See Figure 4).

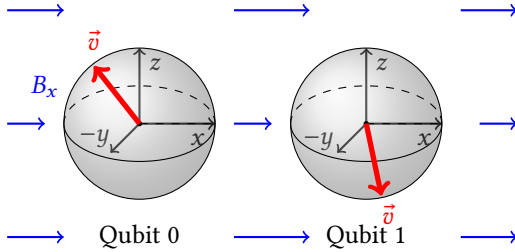


Figure 4. Two-Qubit Ising Model: Phase Transition

The region about the phase transition was a key focus of this project, as the behaviour of the system includes quantum properties.

2.1.2 Four-Qubit Model Now, consider a four-qubit Ising model (See Figure 5). The Hamiltonian for the system becomes:

$$\hat{H} = -J(Z_1Z_2 + Z_2Z_3 + Z_3Z_4) + h(X_1 + X_2 + X_3 + X_4) \quad (3)$$

As expected, this is a more complex system, so its behaviour around the phase transition is less predictable. This project aims to investigate the four-qubit system through computational methods.

2.2 Variational Quantum Eigensolvers

A variational quantum algorithm is an algorithm that uses both quantum and classical components to determine the ground state

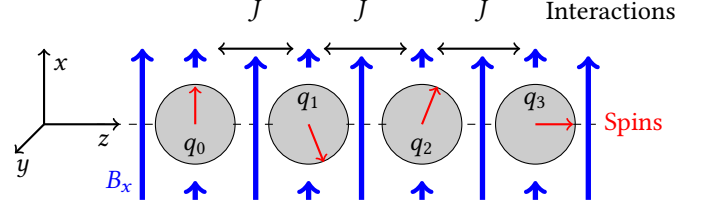


Figure 5. Four Qubit Ising Model

(or lowest) energy of a given physical system. A variational quantum eigensolver (VQE) is a subset of this category that utilizes eigenvalues and eigenvectors directly in its calculations (Watrous, 2025b). The algorithm uses a parameterized quantum circuit, called an ansatz, a corresponding parameterized function, and a classical optimizer backend. To better comprehend the operation of a VQE, a description of each step is given (See Figure 6).

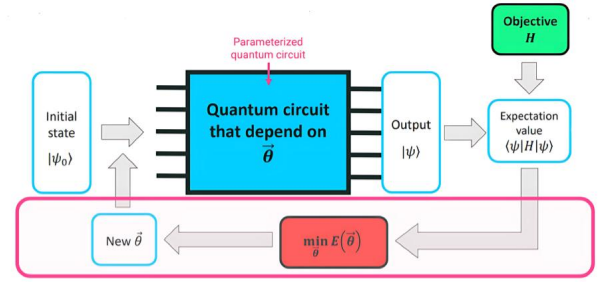


Figure 6. Variational Quantum Algorithm Workflow

2.2.1 Initialize Problem Before the algorithm can run, the problem must be defined appropriately. This involves defining a parameterized function to minimize and initializing the quantum computer in a default state of $|0\rangle$. In the case of the transverse field Ising model, a function known as the "cost function" is utilized.

2.2.2 Prepare Ansatz The ansatz is the parameterized quantum circuit utilized in the VQE. The design of an ansatz determines the efficiency and accuracy of the VQE. Common ansatzes used in Qiskit are: RealAmplitudes, EfficientSU2, QAOA, and TwoLocal (IBM, 2025). The goal of the ansatz is to turn the initial state $|0\rangle$ into some target state $|\psi(\vec{\theta})\rangle$.

2.2.3 Evaluate Cost Function After obtaining some target state $|\psi(\vec{\theta})\rangle$, the cost function is evaluated. In the case of the transverse field Ising model, the cost function is represented by the expectation value (energy) of the system $(\langle\psi|\hat{H}|\psi\rangle)$.

2.2.4 Optimize Parameters Once the expectation value is calculated, the output is transferred to a classical optimizer which determines the minimum of the function and the parameters at that point.

2.2.5 Adjust Parameters and Re-Run These new parameters are then applied to the ansatz and the cycle runs again. These steps repeat until the VQE converges on a particular expectation value (or system energy). This is the given ground state energy of the physical system.

2.2.6 Summary Variational quantum eigensolvers are used for a variety of quantum problems, which limits the specificity of this description. Further, the design of ansatz circuits is an active field of research that continues to evolve as the capabilities of quantum computers are realized. For more information, see Watrous (2025c).

3 Methods

Within this section, a description of the particular methods used within the project are included. A brief summary of the findings is included but conclusive analysis can be found in the Results section.

3.1 Ansatz Comparison

To begin the investigation, the most efficient VQE was determined. The most effective way to change the efficiency of a VQE is to alter the Ansatz it uses. In this investigation, we limited the scope of the ansatz to pre-coded circuits in Qiskit. This included: TwoLocal, QAOA, EfficientSU2, and RealAmplitudes (See Figures 7, 8, 9, 10 for these circuits on a single qubit, respectively). To measure the efficiency of the ansatz, we applied each (via a VQE) to a two-qubit transverse field Ising model. We measured the average number of iterations required to converge to a ground state value and the percent error of the final ground state energy compared to exact, classically-determined solutions. To limit the potential error, each VQE ran 200 times to determine the average number of iterations until convergence. The percent error underwent the same procedure. Further, each ansatz included sixteen parameters to optimize, enabling a more accurate comparison.

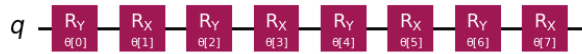


Figure 7. Two Local Ansatz Diagram

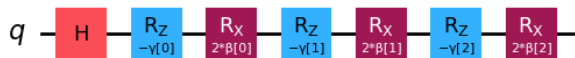


Figure 8. QAOA (Quantum Approximation Optimization Algorithm) Ansatz Diagram

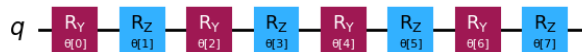


Figure 9. EfficientSU2 Ansatz Diagram



Figure 10. RealAmplitudes Ansatz Diagram

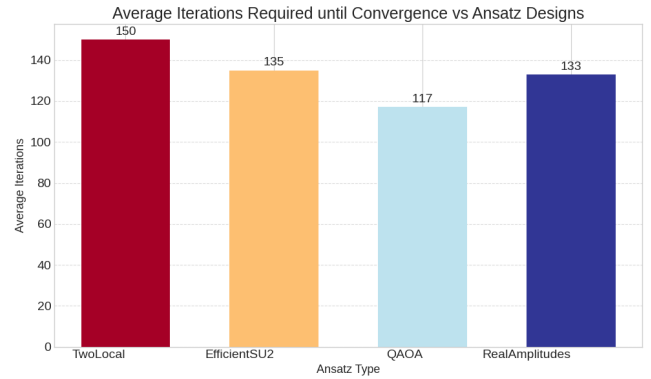


Figure 11. Average Iterations Required for Convergence based on Ansatz Type

3.1.1 Energy Convergence See Figure 11. The results demonstrated that the RealAmplitudes ansatz required the least number of iterations to converge, followed by the QAOA, EfficientSU2, and TwoLocal ansatzes. This indicates that RealAmplitudes requires the least amount of computational energy.

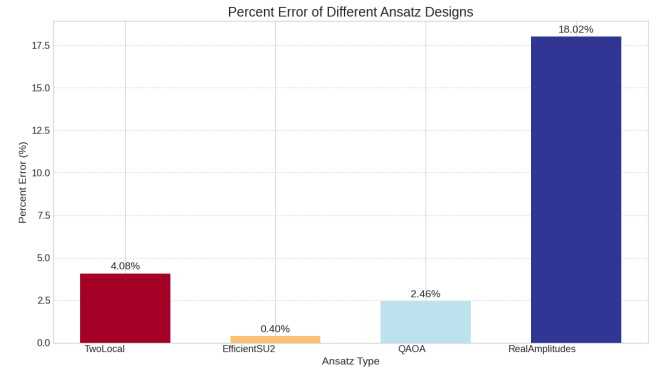


Figure 12. Percent Error between VQE and Classical Solution for Ground State Energy

3.1.2 Percent Error See Figure 12. It was found that RealAmplitudes had the highest percent error, followed by the TwoLocal, EfficientSU2, and QAOA ansatzes. This indicates that the QAOA ansatz has the highest accuracy when compared to the exact values of ground state energy. Precision, accuracy, and computational efficiency are very important when modeling complex physical systems.

Overall, it was determined that the QAOA ansatz was the most optimal ansatz due to the low percent error and relatively low number of iterations required to converge.

3.2 Two-Qubit Ising Model

Once the VQE's efficiency was maximized, those findings were applied to a physical system. Initially, a two-qubit transverse field Ising model was considered. It had the following Hamiltonian Equation:

$$\hat{H} = -JZ_1 \otimes Z_2 + h(X_1 + X_2)$$

where $J = 1$ and $h \in [-2, 2]$. The system was modeled computationally using the Aer Simulator in Qiskit.

3.2.1 Ground State Energy Convergence By utilizing a classical optimization backend entitled "cobyla," the variational quantum eigensolver's convergence was tested at discrete values of the magnetization constant, h . This ensured the coded model was working appropriately by comparing its accuracy to the ideal solutions (calculated using linear algebra methods). The VQE was tested at $h = 0$ and $h = 0.5$ and the results can be seen in Figures 13 and 14.

Ground State Energy Convergence for Two-Qubit Transverse Field Ising Model ($h=0$)

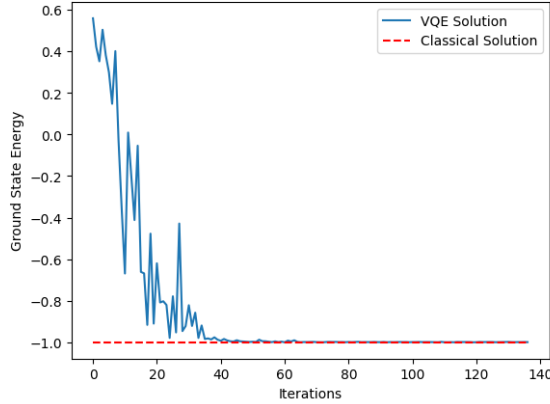


Figure 13. Convergence on Ground State Energy of Two-Qubit Transverse Field Ising Model: Comparison of quantum and classical solutions for $h = 0$

Ground State Energy Convergence for Two-Qubit Transverse Field Ising Model ($h=0.5$)

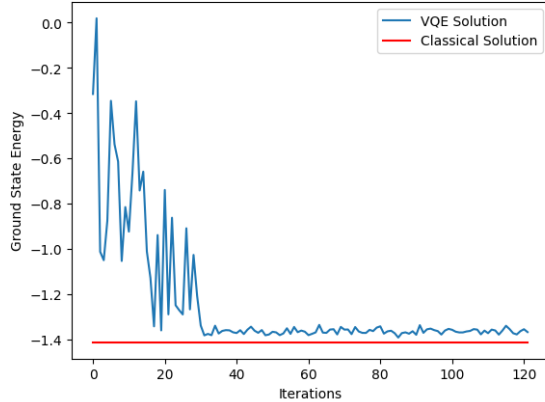


Figure 14. Convergence on Ground State Energy of Two-Qubit Transverse Field Ising Model: Comparison of quantum and classical solutions for $h = 0.5$

It is clear that the variational quantum eigensolver was converging to the correct approximate values. Occasionally, floating point errors became a cause of concern, but the convergence was accurate enough to continue with the exploration.

3.2.2 Ground State Energy with Varying Field After ensuring the VQE was correctly converging to the ground state, the investigation was continued by varying the field constant h and determining its impact on the ground state energy of the system. The graph was predicted to be symmetric about $h = 0$, as the field direction is the only variation between $h \in [-2, 0]$ and $h \in (0, 2]$.

Initially, the step size was arbitrarily set as 0.5 to test the stability of the calculations, but was quickly decreased to 0.05 to yield the following results (See Figure 15).

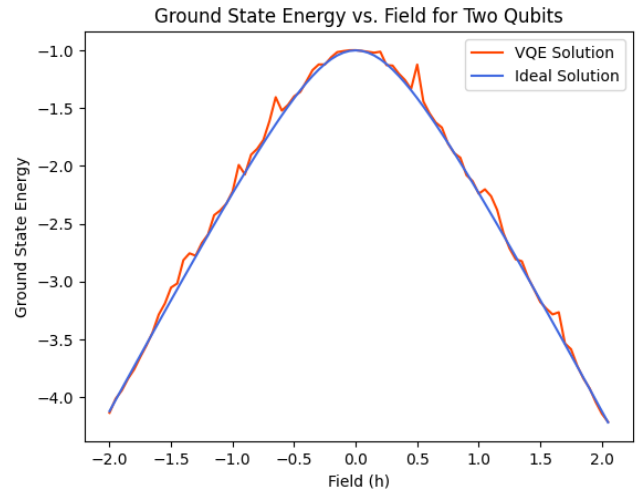


Figure 15. A comparison of quantum and classical solutions in the context of a Two-Qubit Transverse Field Ising Model's Ground State Energy vs. External Field Variation

It is clear that there is some variation between the VQE and the ideal solution, but the general shape is the nearly identical. An initial prediction was proven true, as the graph was symmetric about $h = 0$. However, attention was drawn to the fact that the difference between the ideal and VQE solution varied on either side of $h = 0$. The error was not symmetric (as shown in Figure 16). Further, it is interesting to note that the behaviour about the phase change (where $h \approx J = 1$) is unremarkable, which contrasts with the later findings in the four-qubit model.

3.2.3 Magnetization with Varying Field We extended the investigation by using the values found as the ground state energy to calculate the magnetization of the system along the the Z-axis (See Figure 17). There is also magnetization in the X-axis, but those calculations were outside the scope of this project (as it would requires a different cost function for the VQE).

The step size was 0.22 for these calculations, as the VQE's ability to converge collapsed on smaller step sizes.

One of the most notable aspects of this graph is the accuracy of the VQE compared to the classical solution. They follow each other almost exactly. Once again, the area around the phase transition is unremarkable, which indicates that the simplistic system

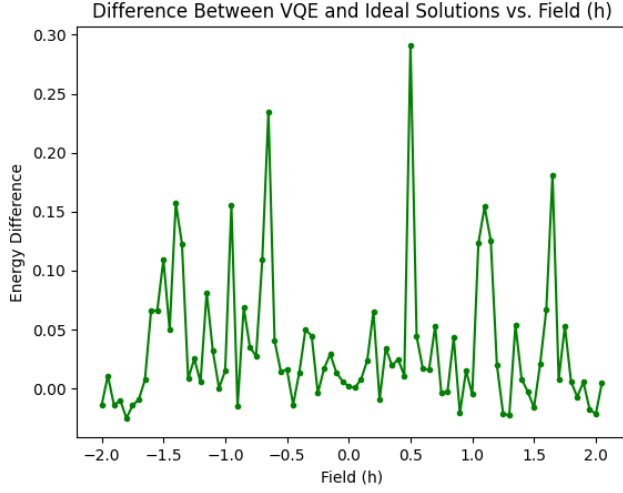


Figure 16. Difference between VQE and Ideal solution for ground state energy of Two-Qubit Transverse Field Ising Model vs. External Field Variation

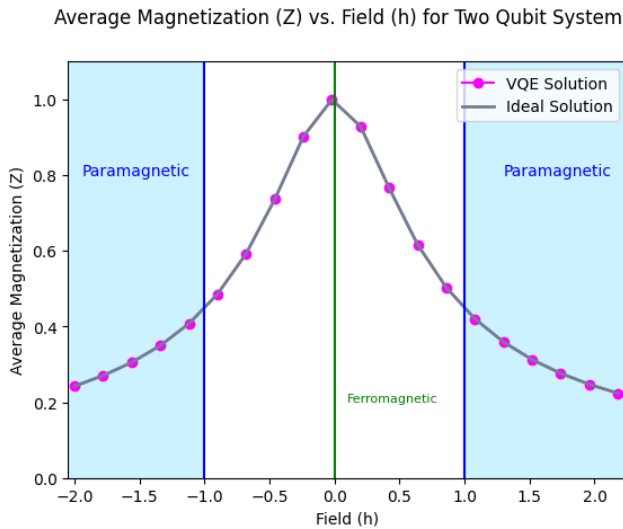


Figure 17. Magnetization in the Z-Axis of a Two-Qubit Transverse Field Ising Model as External Field Varied

does not experience any drastic changes as an external field is applied.

3.3 Four-Qubit Ising Model

Next, a four-qubit transverse Ising Model was considered (See Figure 5). It had the following Hamiltonian Equation:

$$\hat{H} = -J(Z_1 \otimes Z_2 + Z_2 \otimes Z_3 + Z_3 \otimes Z_4) + h(X_1 + X_2 + X_3 + X_4)$$

where $J = 1$ and $h \in [-2, 2]$. The system was modeled computationally using the Aer Simulator in Qiskit.

3.3.1 Ground State Energy Convergence Once again, the stability of the variational quantum eigensolver was checked by ensuring its convergence to the ground state energy of the system (See Figure 18). To avoid redundancy, only the graph for $h = 0$ is included. The VQE was found to converge to the correct ground state energy, within reason.

Ground State Energy Convergence for Two-Qubit Transverse Field Ising Model ($h=0$)

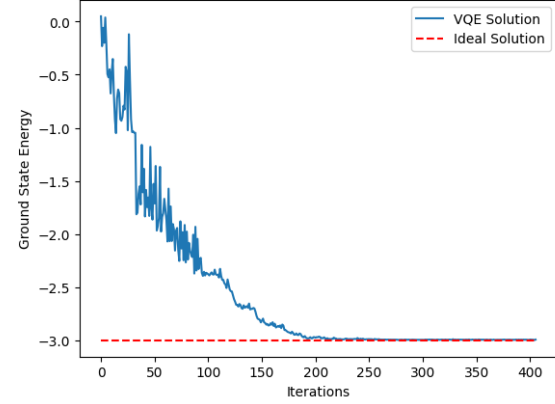


Figure 18. Convergence of variational quantum eigensolver on ground state energy of four-qubit transverse field Ising model

3.3.2 Ground State Energy with Varying Field After ensuring the variational quantum eigensolver was not encountering computational errors in the more complex system, the field constant h was varied (See Figure 19). Again, the graph was predicted to be symmetric about $h = 0$, as the field direction is the only variation between $h \in [-2, 0)$ and $h \in (0, 2]$. An additional prediction expected greater variation between the VQE solution and ideal solution due to the increased complexity of the system.

Initially, the step size was arbitrarily set as 0.5 to test the stability of the calculations, but was quickly decreased to 0.05 to yield the following results (See Figure 19).

Once again, we see that around the phase transition, there is smooth, continuous behaviour. This behaviour indicates the stability of the ground state energy of the system around this point. The energy stays relatively predictable while the magnetization does not, as seen later.

To further improve the accuracy of the VQE solution, the algorithm was run five times for each value of h and the ground state energy result was averaged. This yields the results seen in Figure 20, where the VQE follows the ideal solution more precisely.

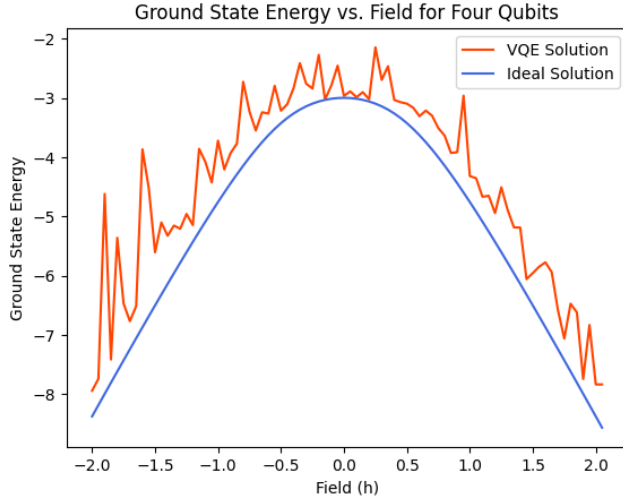


Figure 19. Ground state energy of four-qubit transverse field Ising model as external field varies (One iteration of algorithm)

To visualize the difference in error between the VQE and ideal solution, each was calculated and plotted on the same graph (See Figure 21). It is clear that more iterations are required for accurate convergence of the VQE on more complex physical systems.

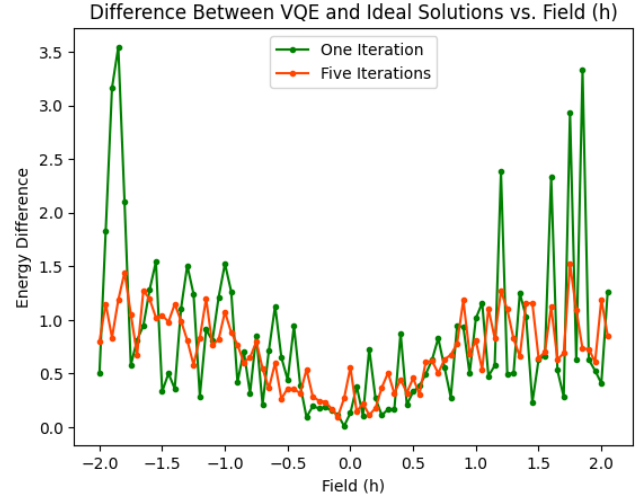


Figure 21. Difference between VQE and Ideal solution for ground state energy of Four-Qubit Transverse Field Ising Model vs. External Field Variation for a single iteration and the average of five iterations of the variational quantum algorithm

3.3.3 Magnetization with Varying Field To continue, the values found as the ground state energy were used to calculate the magnetization of the system along the the Z-axis (See Figure 22).

Once again, the step size was 0.22 for these calculations, as the VQE's ability to converge collapsed on smaller step sizes.

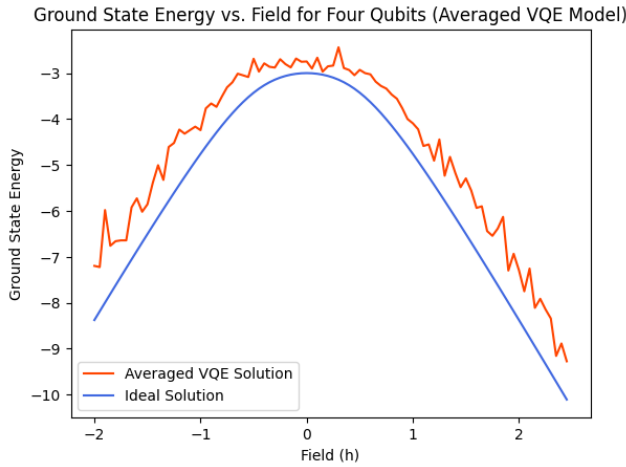


Figure 20. Ground State energy of four-qubit transverse field Ising model as external field varies (Five iterations of algorithm averaged)

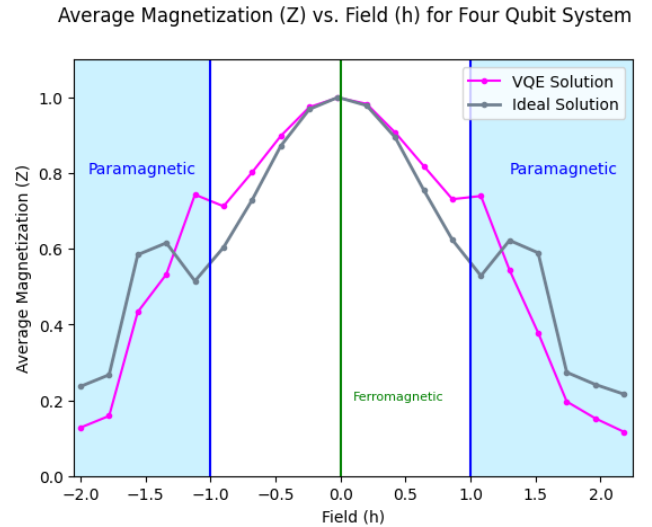


Figure 22. Magnetization in Z-axis of four-qubit transverse field Ising model as external field varies

It is clear that there is much more variation and unpredictability regarding the four-qubit system's magnetization in the Z-axis.

While the odd behaviour is expected to occur near the phase transition (where $h \approx J = 1$), it tends to occur just beyond these points on either end instead. This behaviour demonstrates the stronger impact of the field on the system beyond this critical point. It is also noted that there are more obvious differences between the VQE solution and classical solutions when calculating the magnetization due to the complex nature of the system and inherent functions of our particular VQE.

4 Results

The two-qubit and four-qubit transverse field Ising models were considered in the comparison of a variational quantum eigensolver and classical, linear algebra methods. Both models were one-dimensional (all particles were aligned along the Z-axis), which led to similar results. Preliminary testing of the variational quantum eigensolver ensured its ability to converge to the ground state of the system, within computational reason. Then, consistent results regarding the ground state energy of the system as the external field was varied were determined. This is likely due to the one-dimensional nature of the problem and how only nearest-neighbor interactions were considered. The fundamental assumptions of the model were very similar, which makes the consistent, parabolic nature of the ground state energy graphs under different external field strengths reasonable.

Despite this, the variation between the VQE and ideal ground state energy values was drastically different for each model. The two-qubit model had differences in values in the range of $[-0.05, 0.30]$, while the four-qubit model (averaged over five iterations) had differences in the range of $[0, 1.5]$. It is clear that for more complex systems, more computational power is required for accurate solutions (due to the larger number of iterations the VQE requires to attain reasonably accurate values).

In the investigation of magnetization, however, there was fairly low consistency between the two- and four-qubit models. The two-qubit model showed consistently accurate VQE results (compared to the ideal solution) and unremarkable behaviour about the phase transition, indicating the minimal impact of competing interactions on a small system. However, the four-qubit model demonstrated considerably different values of the VQE and ideal solution, especially about the phase transition. This variability about the phase transition makes it clear that competing interactions (those between two particles and those between a single particle and an external field) have a greater impact on a larger system.

5 Discussion

In the greater context of the quantum computing field, this exploration revealed a key limitation of variational quantum eigensolvers. The phenomenon is known as a "barren plateau." A "barren plateau" occurs when the optimization function reaches a minimum that may not be the global minimum of the function. The optimization function "gets stuck" and misinterprets the value at that point as the global minimum. As a result, the VQE results can vary from the ideal solution, which avoids issues such as the "barren plateau."

Despite these issues, quantum computing power has been shown to scale polynomially with problem size, which makes it more efficient for larger physical models. Classical computing

power scales exponentially with problem size. At a particular problem size, quantum computing becomes more efficient than classical computational methods. Hence, it is safe to conclude that for larger physical systems, quantum computational methods, such as variational quantum eigensolvers, are the more efficient option.

6 Future Research

Future explorations in the field of quantum computation could include a simulation of noise within the variational quantum eigensolver. Including noise would more accurately simulate the conditions of a real quantum computer (instead of the ideal simulator used). Other investigations could focus on other non-Ising models that rely on different interactions between particles. Determining variational quantum eigensolver uses in a variety of models can exemplify its applicability in a multitude of physical scenarios.

Acknowledgements

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